

DATA TASK REPORT

Data Collection & Preprocessing

Time-Aware Explainable ML for Early At-Risk Student Detection (OULAD) — Task 3

DSP391m · Data Science Capstone · Group 1 · FPT University

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Phuc · Binh

Agenda

- **Part 1** — Defining the required data (Son)
- **Part 2** — Data collection method (Phuc)
- **Part 3** — Data cleaning (An)
- **EDA block** — Exploratory data analysis (Duc)
- **Part 4** — Transformation & standardisation (Binh)
- **Part 5** — Train/test split & leakage control (Khoa)

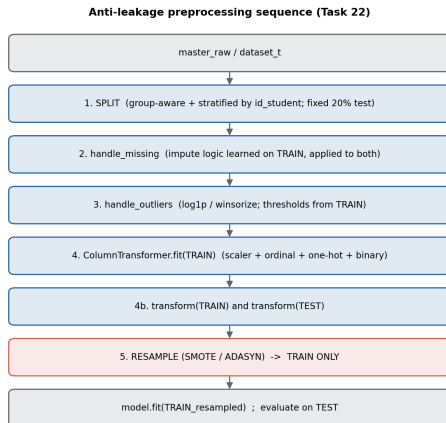
⇒ *Every item follows the same frame: What — Why — Result.*

The Big Picture — Our Data Pipeline

What 7 raw OULAD tables → merge → `master_raw` ($32,593 \times 33$) → cut at 6 time checkpoints → leakage-safe preprocessing → train/test.

Why An “anchor” slide so we always know which pipeline step we are in.

Result 6 datasets `dataset_t10...t100` + a fixed 20% test set; **19/19** leakage tests pass.



Anti-leakage preprocessing sequence

PART 1

Defining the Required Data

Presenter: Son

Problem & Objective

What Detect **early** the students at risk of poor performance on the VLE, and **explain** every prediction.

Why Three gaps in prior work: model opacity, late prediction, and class imbalance.

Result The data must serve RQ1 (early prediction), RQ2 (explanation stability), RQ3 (imbalance).

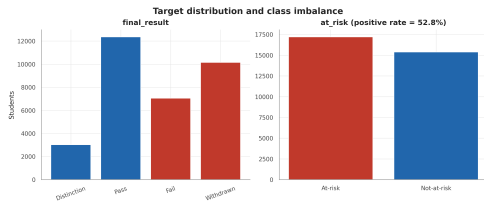
Target Variable (`at_risk`)

What Binary classification: `at_risk` derived from `final_result`; at-risk (1) = Fail + Withdrawn, not-at-risk (0) = Pass + Distinction.

Why Group by “does this student need intervention” — exactly the early-warning goal.

Result Real distribution, not 68/32.

► **Key:** at-risk **52.8%** (17,208) vs not-at-risk **47.2%** (15,385).



Target distribution & class imbalance

Three Input Feature Groups

What Three feature groups: **Demographic** (gender, region, education, imd_band, age_band, disability); **Engagement/VLE** (clickstream: total/typed clicks, active days); **Assessment** (submission scores, count, not-submitted flag).

Why Demographics = context & fairness; engagement = early behavioural signal; assessment = evidence of performance.

Result A feature set split into 3 clear groups $\Rightarrow$ group-wise SHAP/LIME interpretation.

Observation Unit, Scope & Exclusions

Unit One row = one student-module-presentation (`code_module` × `code_presentation` × `id_student`).

Scope 32,593 records · 22 module-presentations · 7 tables · Open University 2013–2014.

Exclude Drop identifiers; **never** use `date_unregistration` as a feature; drop the raw `final_result`.

Why Fixing the unit and excluding leaky columns early prevents leakage and row duplication.

PART 2

Data Collection Method

Presenter: Phuc

Source & Data Type

What Source: Open University Learning Analytics Dataset (OULAD), Kuzilek et al. (2017); type: public **secondary** data, anonymised at source, CC-BY 4.0.

Why Ethics satisfied by correct citation; no sensitive personal data is processed.

Result Legal, reproducible, no primary collection needed — fits an academic setting.

Collection & the 7 Tables

What Download 7 CSVs → `data/raw/` → verify integrity with MD5 (`data_manifest.txt`). Tables: `studentInfo`, `courses`, `studentRegistration`, `studentVle` (**10,655,280** rows), `vle`, `assessments`, `studentAssessment`.

Why Separate light base tables from the heavy clickstream → chunked processing (500k rows) avoids memory blow-up.

Result Raw data “frozen” + checksummed ⇒ everyone uses exactly one version.

Relationships Between Tables

What Primary keys:

- (code_module, code_presentation, id_student) — studentInfo ↔ registration ↔ VLE-agg ↔ performance.
- id_site — studentVle ↔ vle (activity type).
- id_assessment — studentAssessment ↔ assessments (weight/deadline).

Why Knowing the keys is the precondition to merge **without duplication** (validate='many_to_one').

Result A relational schema — the basis for the Part-3 merge.

Source Evaluation: Fit · Pros · Limits · Risks

What Fit: all 3 feature groups + timestamps → enables time-aware prediction; 32k is large enough. Pros: international benchmark, anonymised, reproducible, free.

Why Limits: secondary, single institution; clicks are counts only (no content); 2013–2014 cohorts.

Result Risks: mild imbalance · structural missingness · strong right-skew · multicollinearity · temporal leakage — all addressed in Parts 3–5.

PART 3

Data Cleaning

Presenter: An

Raw-Data Overview

What After merging: 32,593 rows $\times$ 33 columns; audit data types & structure.

Why A “general check-up” before intervening — know the scale and variable kinds.

Result A rows–columns–type summary as the baseline for cleaning.

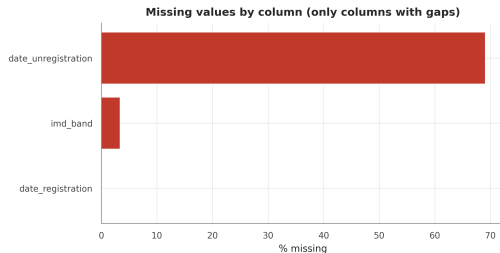
Missing Data — Detection & Strategy

What Only 3 columns missing; missing scores = “not submitted yet” → fill 0 + `not_submitted` flag.

Why Each mechanism (MCAR/MAR/MNAR) is handled differently; “not submitted” is **signal**, not noise.

Result After handling: **0** missing in every feature column.

Column	#	Handling
date_unregistration	22,521	drop (structural)
imd_band	1,111	fill “Unknown”
date_registration	45	train median



Missing values by column

Duplicates & Format Standardisation

What Duplicates: check the student-module-presentation key → `drop_duplicates` → 0 duplicate keys.

What Types: catalogue numeric / ordinal / nominal / binary / indicator (`VARIABLE_TYPES`).

What Standardise: strip text; reconcile dictionaries (region 13, education 5, imd_band 10, age_band 3, gender/disability 2).

Why Correct types enable correct encoding in Part 4; standardising prevents 'Y' ≠ 'Y'.

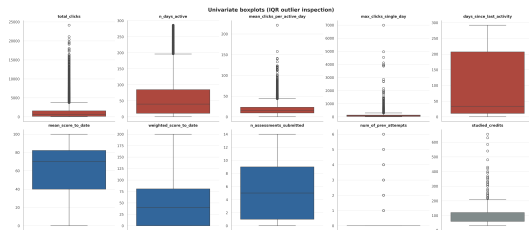
Result A dataset consistent in format & type.

Outliers — Detection

What Logical checks ($\text{score} \in [0, 100]$) + IQR-rule detection (`log_outliers`).

Why Clickstream is strongly right-skewed: `clicks_resource` skew ≈ 34.7 , kurtosis $\approx 2,125$; `max_clicks_single_day` max = 6,988.

Result A 7-variable outlier table — handed off Binh $\rightarrow$ Duc.



Univariate boxplots (IQR inspection)

Outliers — Handling

What Per-variable strategy: `log1p` for 12 strongly-skewed click features · winsorise (clip 1% each tail) for moderately skewed · none for naturally bounded.

Why **No row is deleted:** an at-risk student's extreme values (e.g. many re-takes) are signal to keep.

Result Every numeric feature is “cleaner” in distribution while all 32,593 rows remain.

Merge & Final Clean Dataset

What Merge: `studentInfo (base) ← left join ← registration ← engagement ← performance ← courses`.

Why A before/after log proves no duplication, no loss (`validate='many_to_one'`).

Result `master_raw = 32,593 × 33` — clean, one row per student–module–presentation.

Step	rows_before	rows_after
studentInfo (base)	32,593	32,593
+ registration	32,593	32,593
+ engagement	32,593	32,593
+ performance	32,593	32,593

PART

Exploratory Data Analysis

Presenter: Duc

EDA Objectives

- Understand the data: 32,593 students, 33 variables (demographic · engagement · assessment).
 - Check quality: missing data, types, outliers.
 - Find features that separate at-risk vs not-at-risk students.
 - Serve two decisions: **feature selection & leakage control**.
- **Key:** 32,593 records · 33 columns · 3 feature groups.

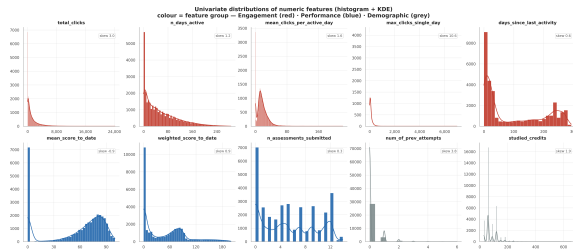
Univariate Analysis

What Histogram + KDE and boxplots per numeric feature.

Why Most clickstream features are **strongly right-skewed**; heavy tails are **real** outliers (super-active students), not removed.

Result $\Rightarrow$ log transform + standardisation; use non-parametric tests.

► **Key:** clicks_resource skew **34.71** · kurtosis **2,125.45**.



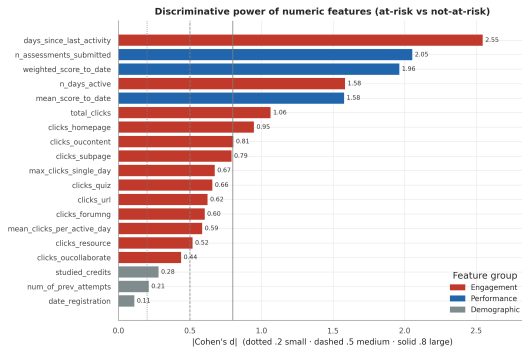
Bivariate Analysis (vs at-risk)

What Compare the two classes with **Cohen's d** + Mann–Whitney (BH-corrected).

Why At $n \approx 32k$ p-values are tiny $\rightarrow$ rank by **effect size**, not p.

Result Top: days_since_last_activity $d=2.55$ ·
n_assessments_submitted 2.05 ·
weighted_score_to_date 1.96.
19/19 features significant.

► **Key:** days_since: 171 vs 14 days ·
n_assessments: 2.3 vs 8.6.



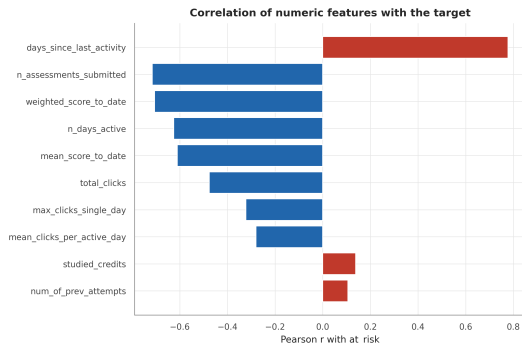
Multivariate — Correlation & Leakage Check

What Pearson (linear) + Spearman (monotonic) heatmaps; correlation with the label; leakage check.

Why Leakage is a **mandatory** step — a feature with $|r| \geq 0.95$ vs the label is suspicious.

Result Top $|r|$ with label: `days_since` $+0.78$ · `n_assessments` -0.72 · `weighted_score` -0.71 . **No feature $|r| \geq 0.95 \Rightarrow$ no leakage.**

► **Key:** $\max r(\text{label}) = 0.78$ · leakage ≥ 0.95 : 0 · Cramér's $V \leq 0.15$.



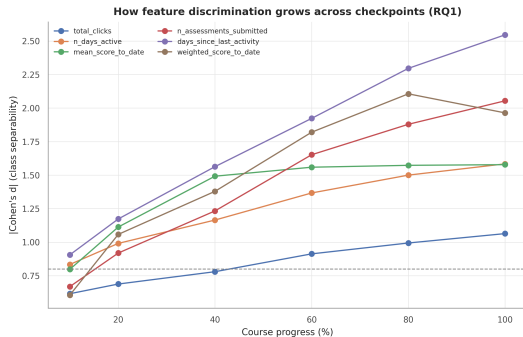
Time-Aware Analysis & EDA Conclusion (RQ1)

What Measure Cohen's d per feature at 6 checkpoints 10%→100%; signal grows steadily.

Why First to reach $d \geq 0.8$:
`n_days_active` and `days_since_last_activity` from 10%; scores/submissions from 20%.

Result Early prediction feasible from ~10–20% of the course; prioritise engagement + assessment features.

► **Key:** `weighted_score` d: 0.61→1.96 ·
`n_assessments`: 0.67→2.05 (10%→100%).



PART 4

Transformation & Standardisation

Presenter: Binh

Encoding Categorical Variables

What Four strategies by variable nature (see table).

Why Wrong encoding (one-hot for an ordinal) **destroys** the order information.

Result Text → numeric; readable by both model and XAI.

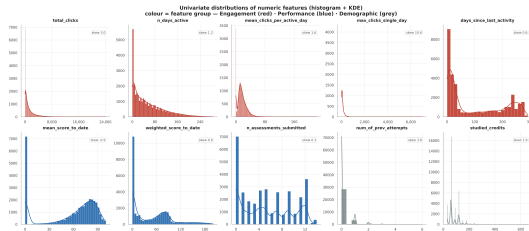
Kind	Variables	Method
Ordinal	highest_education, imd_band, age_band	OrdinalEncoder
Nominal	region, code_module, code_presentation	OneHotEncoder
Binary	gender, disability	BinaryEncoder (0/1)
Indicator	not_submitted	passthrough

Skew Transformation ($\log_{1p}$)

What Apply $\log_{1p}$ to the strongly right-skewed click features (from Part 3).

Why Reduce heavy tails $\rightarrow$ stabilise linear/distance models; evidence: skew ~ 13 – 35 in EDA.

Result Click features become more balanced, no rows lost.



Right-skew $\rightarrow \log_{1p}$

Scaling (StandardScaler)

What StandardScaler brings numeric features to mean 0, std 1.

Why Total clicks (0–thousands) overwhelm scores (0–100) → mandatory for Logistic Regression & ANN.

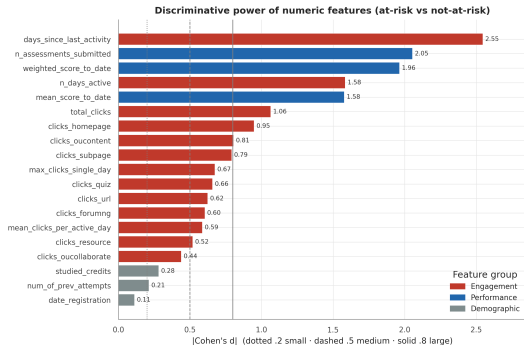
Result Fit on **train only**, transform both train/test (`scaler.mean_` computed from train).

Feature Creation & Significance

What Engagement: total_clicks,
n_days_active, $8 \times \text{clicks_}<\text{type}>$,
max/mean clicks,
days_since_last_activity.
Assessment: mean_score_to_date,
weighted_score_to_date,
n_assessments_submitted,
not_submitted.

Why Each feature carries a hypothesis:
“long inactivity / missed
submissions $\Rightarrow$ higher risk”.

Result Validated by |Cohen's d|:
days_since 2.55 · n_assessments
2.05 · weighted_score 1.96.



Class Imbalance & Plan

What Check: at-risk 52.8% / not-at-risk 47.2% $\rightarrow$ imbalance ratio 1.12 (mild).

Why Missing an at-risk student is the costliest error $\rightarrow$ headline metrics are **PR-AUC & recall** on the at-risk class.

Result RQ3 plan: compare no-resampling / class-weight / SMOTE / ADASYN; resample on **train only**.

PART 5

Train/Test Split & Leakage Control

Presenter: Khoa

Purpose, Ratio & Split Method

What Ratio 80% train / 20% test (`TEST_SIZE = 0.2`).

What Method: `StratifiedGroupKFold` — stratify by `at_risk` + group by `id_student`.

Why A student spans several module-presentations $\Rightarrow$ group by `id_student` so the same person is never in both train and test (group leakage).

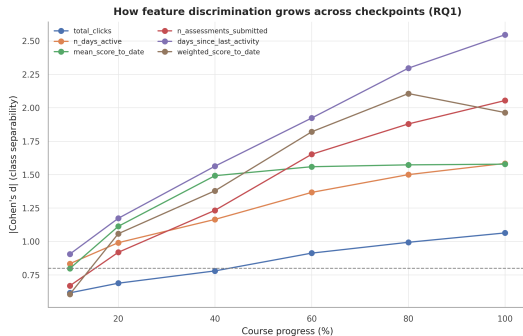
Result Test $\approx$ 6,489 rows / 5,756 students $\cdot$ train $\approx$ 26,104 rows.

Leakage Control — Two Axes

What Time axis: `cut_at_checkpoint()` keeps only events $\leq$ the checkpoint day; `cutoff = round(length  $\times$  t/100)`. Feature axis: every learner (encoder, scaler, resampler) is **fit on train only**.

Why Mirror exactly “the information available at prediction time” — no peeking into the future.

Result 6 datasets `dataset_t10...t100`, identical student list; fixed at-risk 52.8% across checkpoints.



Discrimination grows with time (RQ1)

Correct Preprocessing Order (train→test)

What Order: split train/test → impute → outliers → encode/scale → resample.

Why Parameters (median, winsorise thresholds, mean/std, encoder categories) learned from train → applied to test. Fitting before the split leaks test into train ⇒ fake results.

Result Label distribution after split: train 0.53 / test 0.52 — gap ≤ 0.02 , stratification holds.

Save & Re-Verify — Leakage Blocked

What Save: `test_student_ids.csv` (committed) + train/test parquet;
`split_report.csv`.

What Re-check: 0 students overlapping train/test; no type/column drift; balanced label.

Why A fixed test, reused identically across all 6 checkpoints $\Rightarrow$ 6 **comparable** performance points.

► **Key:** `tests/test_leakage.py`: **19/19** tests pass — automated evidence of no leakage.

PART

Conclusion

Presenter:

Task 3 — Summary

Item	Our result
1. Define data	binary at_risk 52.8%; 3 groups; 32,593 student–module–presentation
2. Collection	OULAD CC-BY 4.0; 7 tables; 10.6M clicks; MD5-verified
3. Cleaning	0 duplicates; log1p+winsorise, 0 rows dropped; master 32,593×33
4. Transform	Ordinal+OneHot+Binary; StandardScaler; d up to 2.55; RQ3 plan
5. Train/test split	80/20 group+stratified; 6 checkpoints; 19/19 tests pass

⇒ All five Task-3 items completed with concrete artifacts and figures.

Outputs & Next Steps

Outputs `master_raw` + 6 datasets `dataset_t10...t100` + a fixed test set + a reproducible pipeline (seed=42, MD5, Restart & Run All).

Next Benchmark models per checkpoint (RQ1) → handle imbalance (RQ3) → SHAP/LIME & explanation stability (RQ2).

Thank you — questions welcome!